ONTOLOGY IN APPLICATION: USE CASES

LGP-ENIT-DIDS TEAM

H.KARRAY

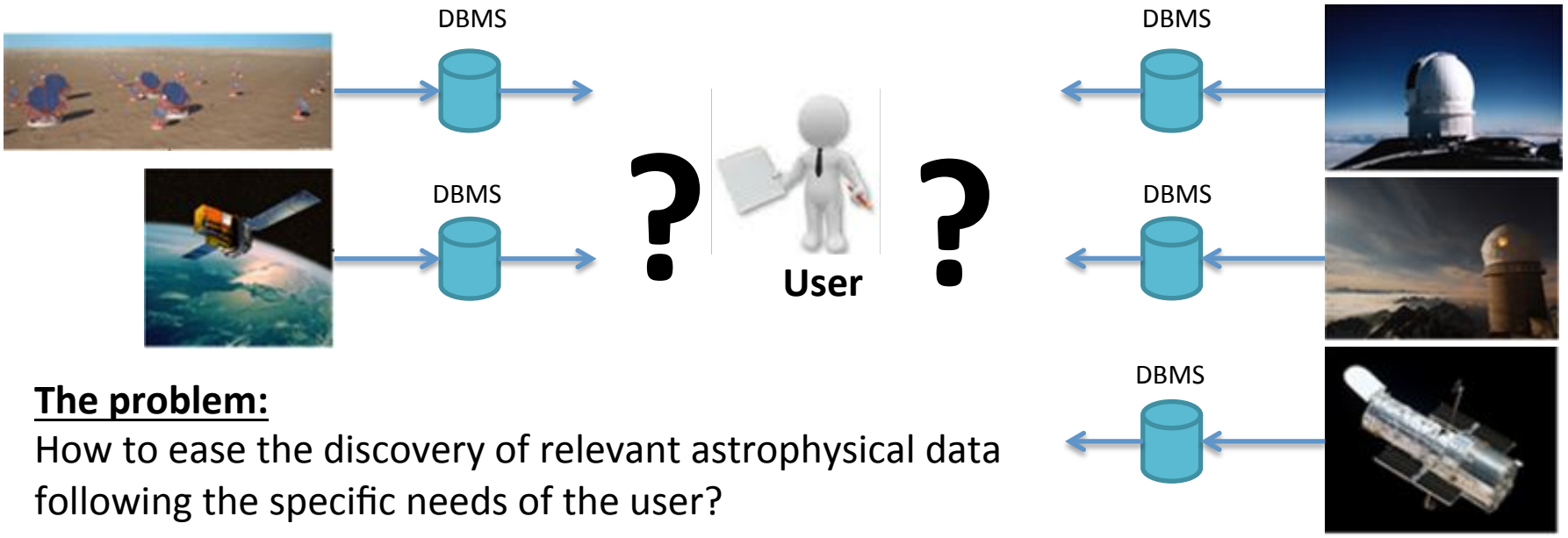
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ASTROPHYSICAL DATA SEARCH USE CASE

ASON: An OWL-S based ontology for astrophysical services. T.Louge, MH. Karray, B. Archimède, J.Knödlseder. (2018) Astronomy and Computing. Vol 24.

CASAS: A tool for Composing Automatically and Semantically Astrophysical Services. T.Louge, MH. Karray, B. Archimède, J.Knödlseder. (2017) Astronomy and Computing Vol 20.

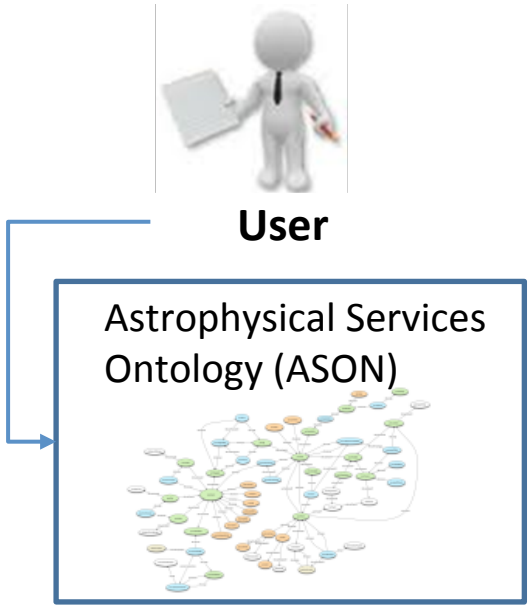
Bringing a services ontology for astrophysical services discovery and composition



The problem:
How to ease the discovery of relevant astrophysical data following the specific needs of the user?

The idea:
It is necessary to provide a standardized way of describing and accessing data, without modifying by any means the existing services proposed by data providers.

The solution we proposed:
ASON, that describes both the domain and the technical aspects of heterogeneous data provider services. Querying ASON does not require any knowledge of predefined keywords, neither any technical proficiency.



Bringing a services ontology for astrophysical services discovery and composition







DBMS
VO Service
DBMS
VO Service
DBMS
VO Service
DBMS
VO Service
DBMS
VO Service

Datamodel n°1
Datamodel n°2
Datamodel n°3

Non-VO compliant services

?

Astrophysical Services Onology (ASON)

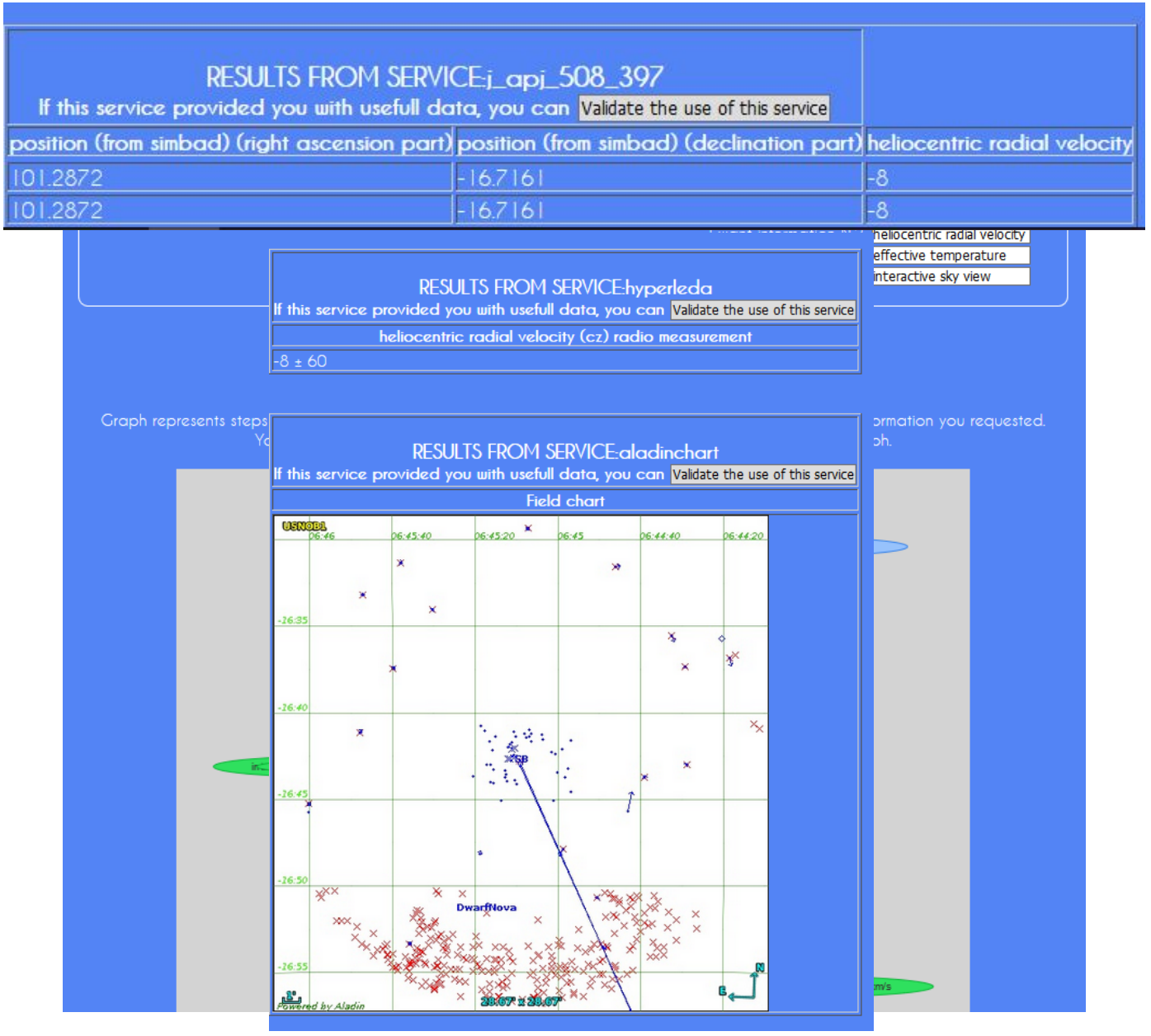





User

Which keywords? Which registries ?

CASAS system interface



DECISION SUPPORT FOR ECOLABELING

Xu, D., Karray, M. H., & Archimède, B. (2018). A knowledge base with modularized ontologies for eco-labeling: Application for laundry detergents. *Computers in Industry*, *98*, 118-133.

Xu, D., Karray, M. H., & Archimède, B. (2017). A semantic-based decision support platform to assist products' eco-labeling process. Industrial Management & Data Systems, 117(7), 1340-1361.

Ecolabeling

- **Issues**
  - Heterogenous, and multi-source criteria documentations with no-computable format .
  - Tedious manual process of product evaluation for experts.

- **Challenges**
  - Achieve an appropriate eco-labeling knowledge representation.
  - Computerize eco-labeling process.

- **Proposed solution**
  - Modular Ontology Knowledge base about ecolabels products and Criteria.
  - Proposition of ontology based decision support system for product's evaluation.

- # ECOLABELING CRITERIA





Amendment ?

ISO ?

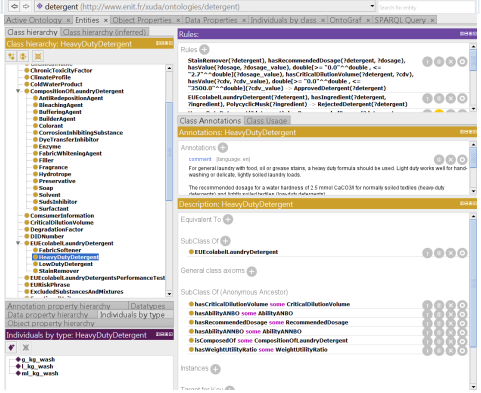


EU Eco-label criteria documents

KB

Translate

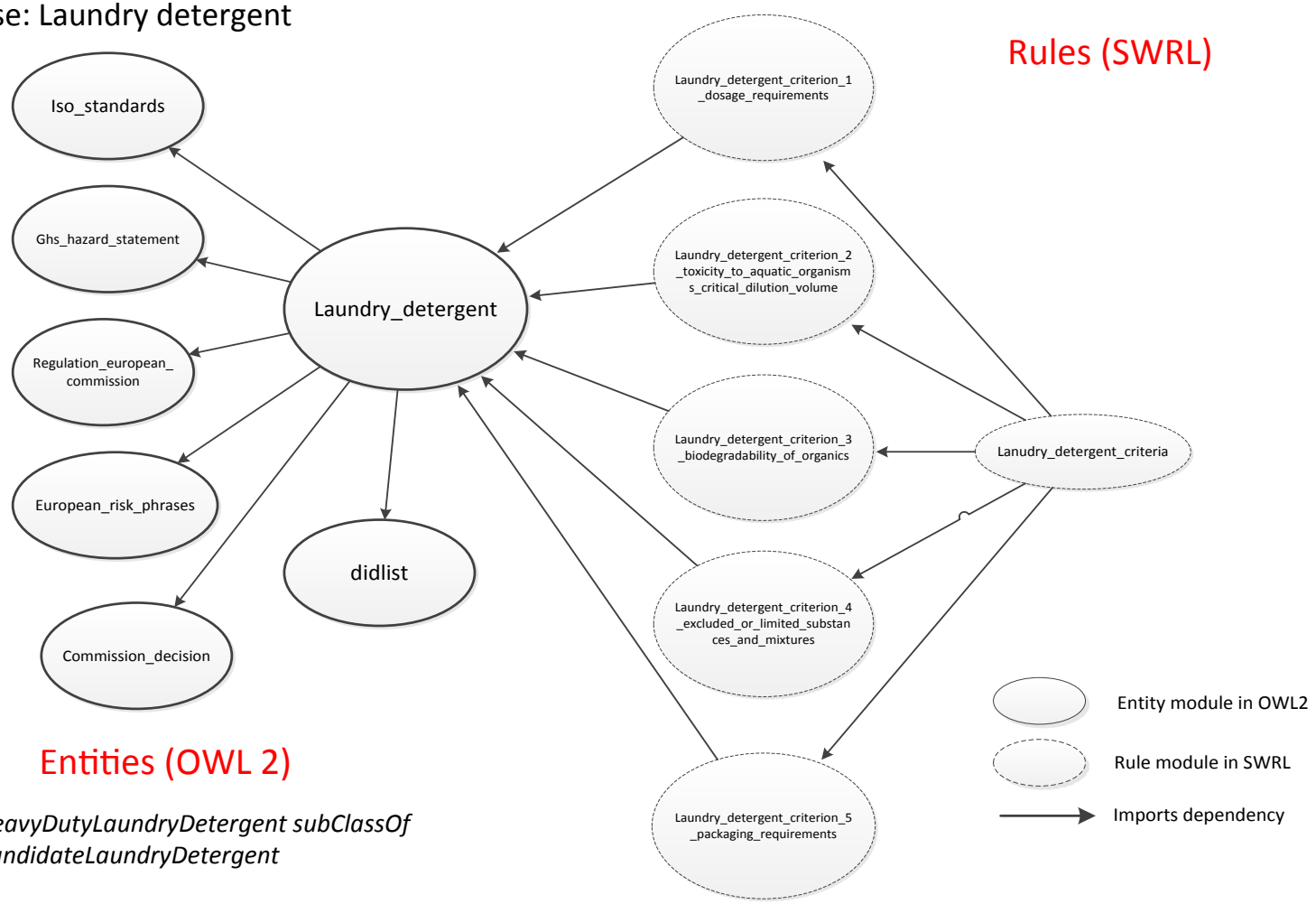
Modeling



OWL 2 Ontology and SWRL Rules

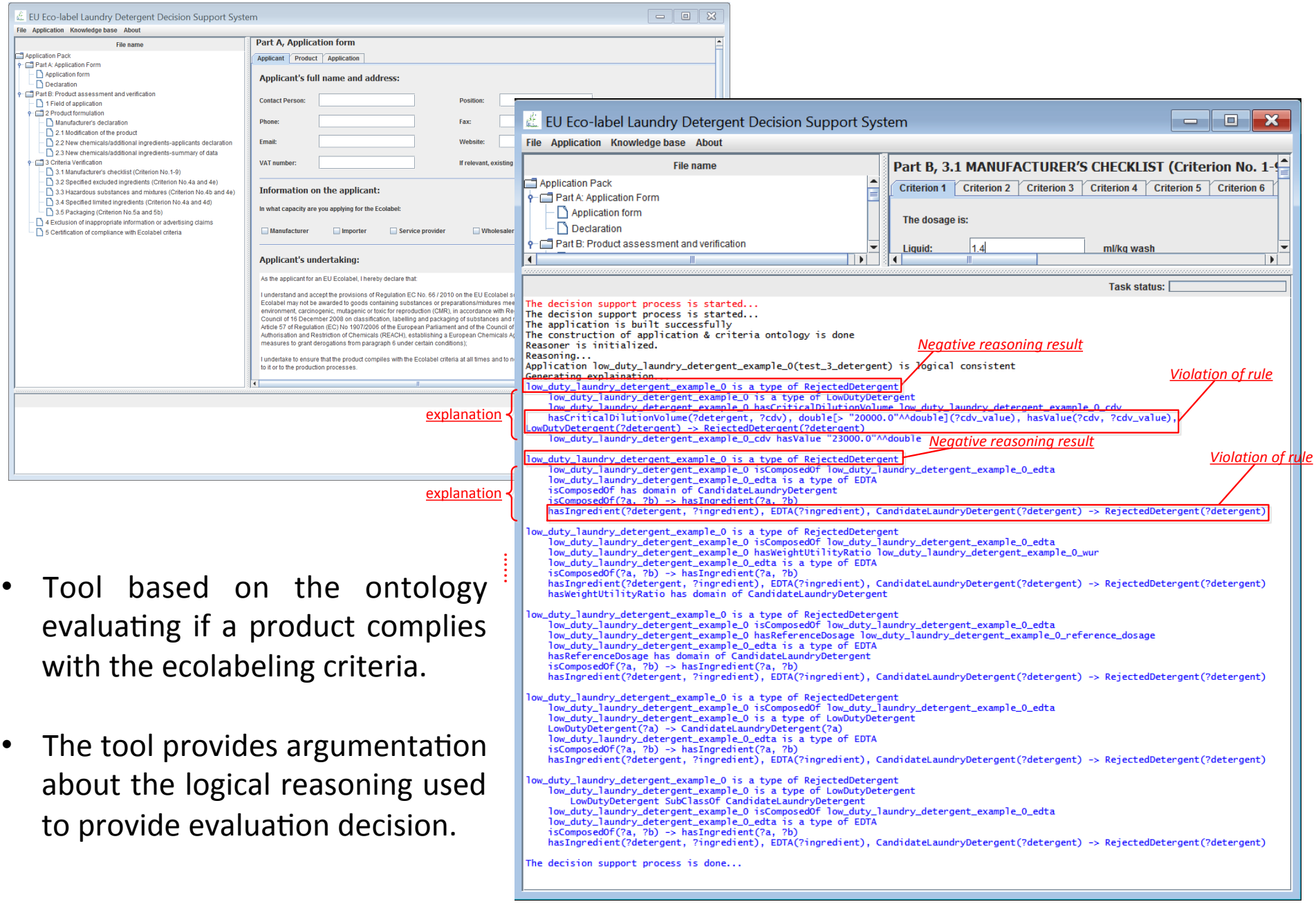
Modularization design and development of the ontology knowledge base taking into account the evolution of eco-labeling criteria

Use case: Laundry detergent



Entities (OWL 2)

e.g. *HeavyDutyLaundryDetergent subClassOf CandidateLaundryDetergent*



- Tool based on the ontology evaluating if a product complies with the ecolabeling criteria.
- The tool provides argumentation about the logical reasoning used to provide evaluation decision.

TRANSPORTATION PLANNING USE CASE

Memon, M. A., Karray, M. H., Letouzey, A., & Archimède, B. (2017). Semantic transportation planning for food products supply chain ecosystem within difficult geographic zones. *Industrial Management & Data Systems*, *117*(9), 2064-2084.

- **Issues**
  - Geographic zones difficult to access.
  - Need to use small tracks.
  - Different kinds of food products.
  - Many constraints: transport conditions, product foods storage requirements, food products association in the same track, etc.











- **Challenges**
  - Grouping food products together while respecting hygienic requirements.
  - Optimise the exploitation of limited transportation means.
  - Collaborate small carriers.



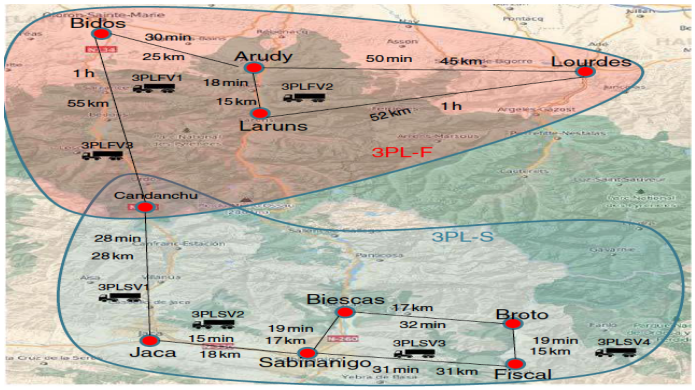






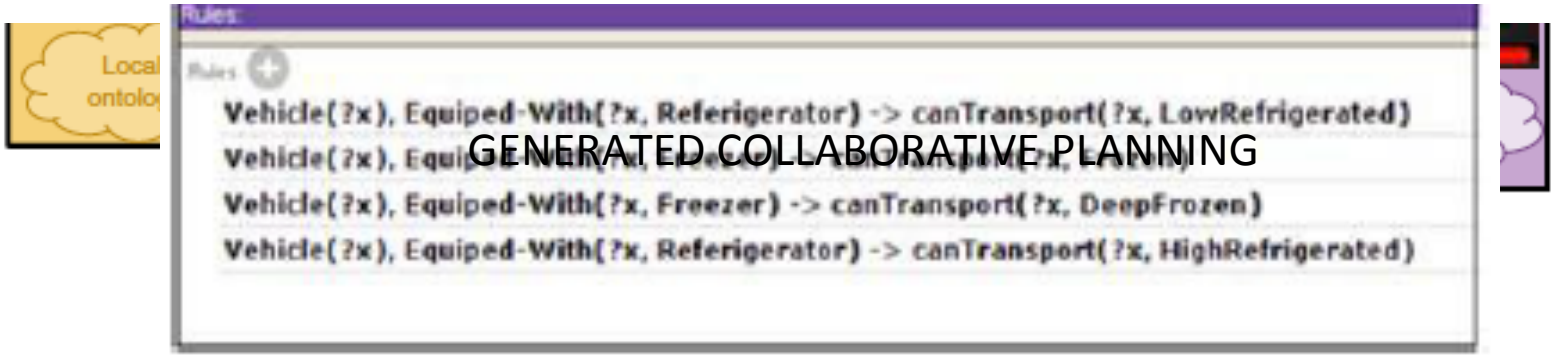


- **Proposed solution**
  - An ontology was set up describing the transport of food products.
  - Market place platform for collaborative food product transportation.



Efficient collborative planing resepcting all differents kind of requirmenets

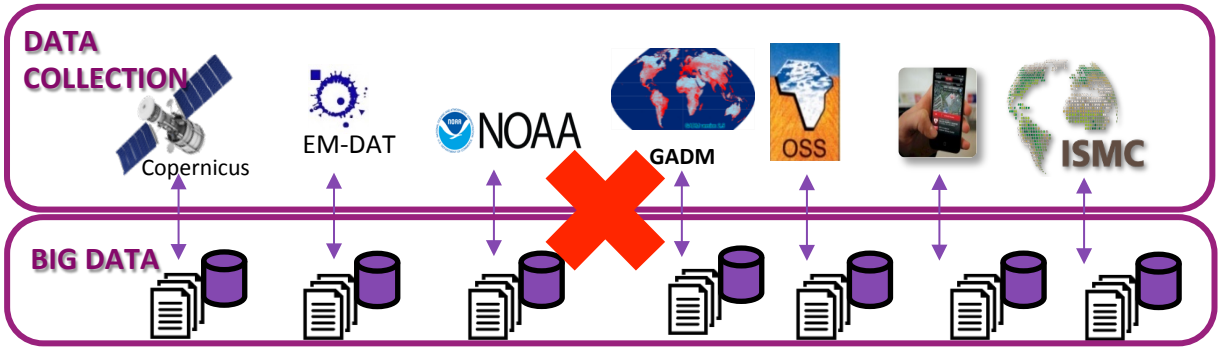




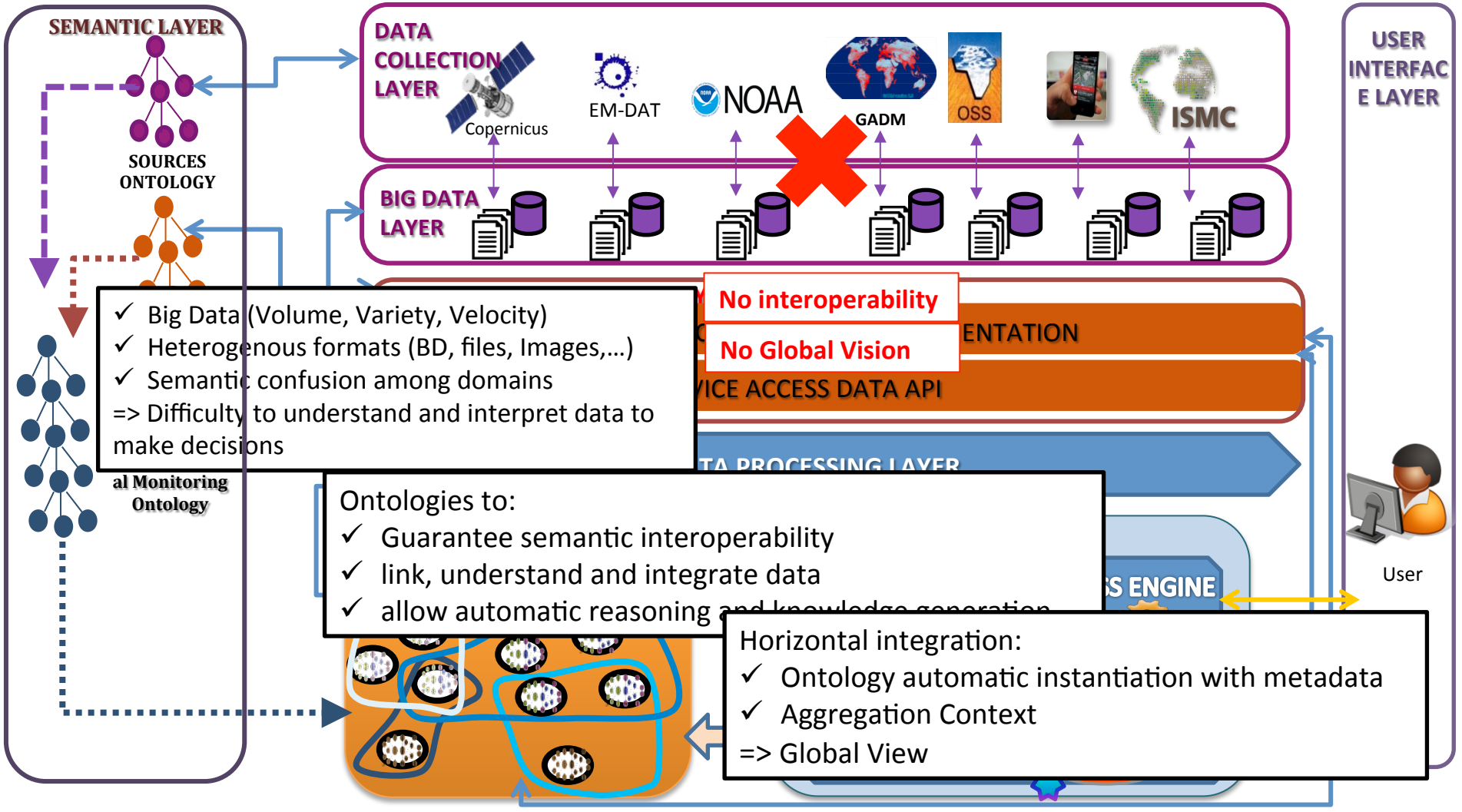
GENERATED COLLABORATIVE PLANNING

SEMANTIC LINKING OF EARTH OBSERVATION DATA USE CASE

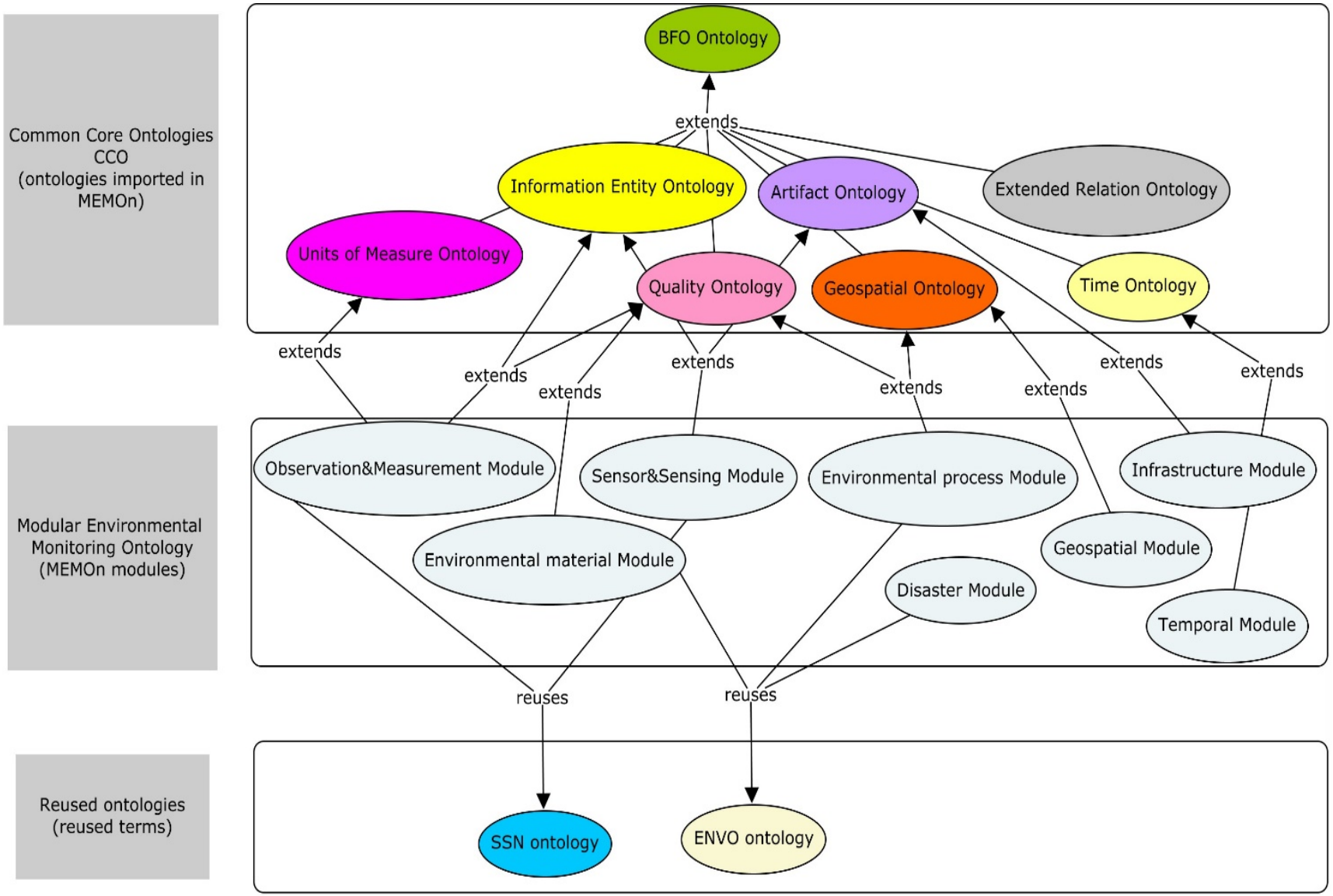
An ontology-based monitoring system for multi-source environmental observations. M. Masmoudi, S. Ben Abdallah Ben Lamine, H. Baazaoui Zghal, MH Karray, B. Archimede. 22nd International Conference on Knowledge-Based and Intelligent Information & Engineering Systems. Proceedings of KES 2018.



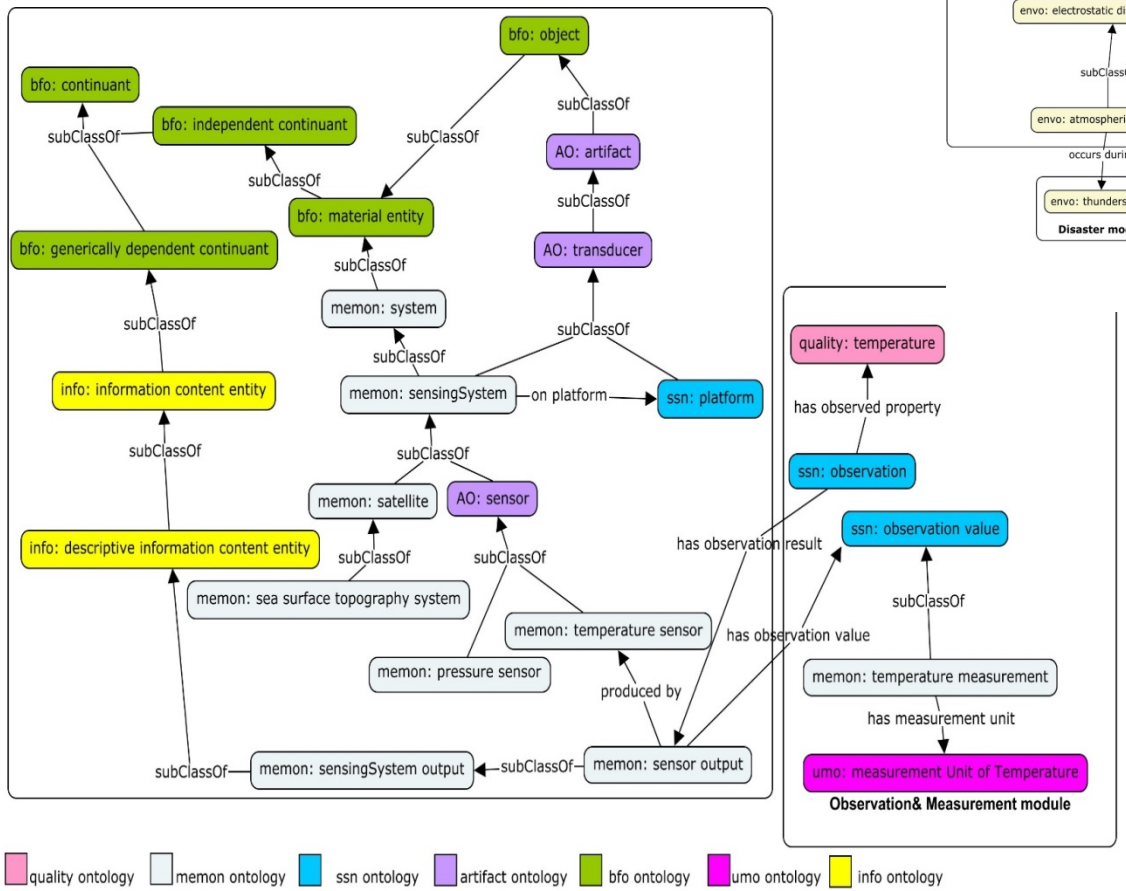
- Issues
  - Heterogenous, massive and multi-source data.
  - Observation and monitoring systems not interoperable.
  - Unpredictable growth of natural disasters.
- Challenges
  - Semantic interoperability between data.
  - Data integration and linking to create a global view and better understand natural phenomena.
  - Generate predictions from massive and multi-source observations.
- Proposed solution
  - A semantic data integration approach for multi-source and big data applied to natural disaster prediction

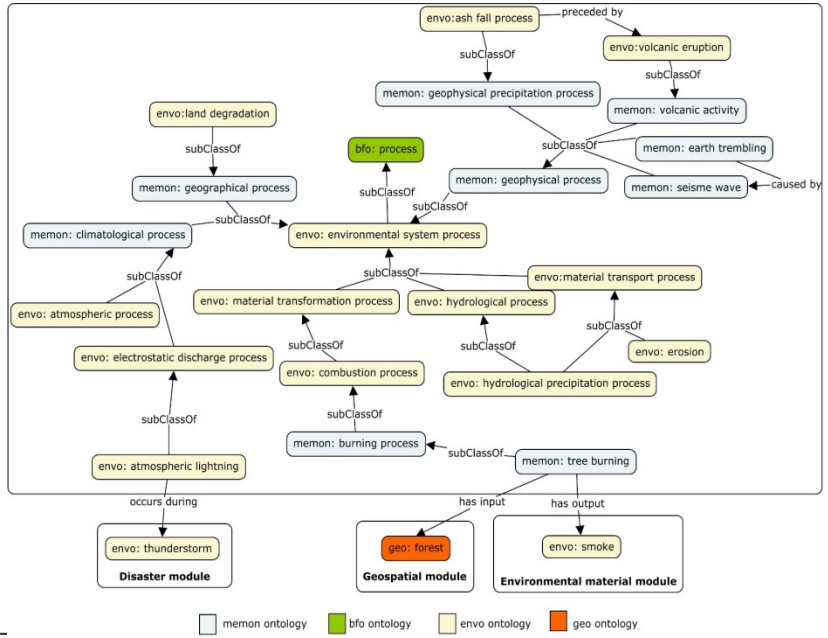

SEMANTIC LAYER
SOURCES ONTOLOGY
al Monitoring Ontology
DATA COLLECTION LAYER
Copernicus
EM-DAT
NOAA
GADM
OSS
ISMC
BIG DATA LAYER
No interoperability
No Global Vision
ENTATION
VICE ACCESS DATA API
TA PROCESSING LAYER
S ENGINE
USER INTERFAC E LAYER
User
✓ Big Data (Volume, Variety, Velocity)
✓ Heterogenous formats (BD, files, Images,…)
✓ Semantic confusion among domains
=> Difficulty to understand and interpret data to make decisions
Ontologies to:
✓ Guarantee semantic interoperability
✓ link, understand and integrate data
✓ allow automatic reasoning and knowledge generation
Horizontal integration:
✓ Ontology automatic instantiation with metadata
✓ Aggregation Context
=> Global View

Modularization of the ontology

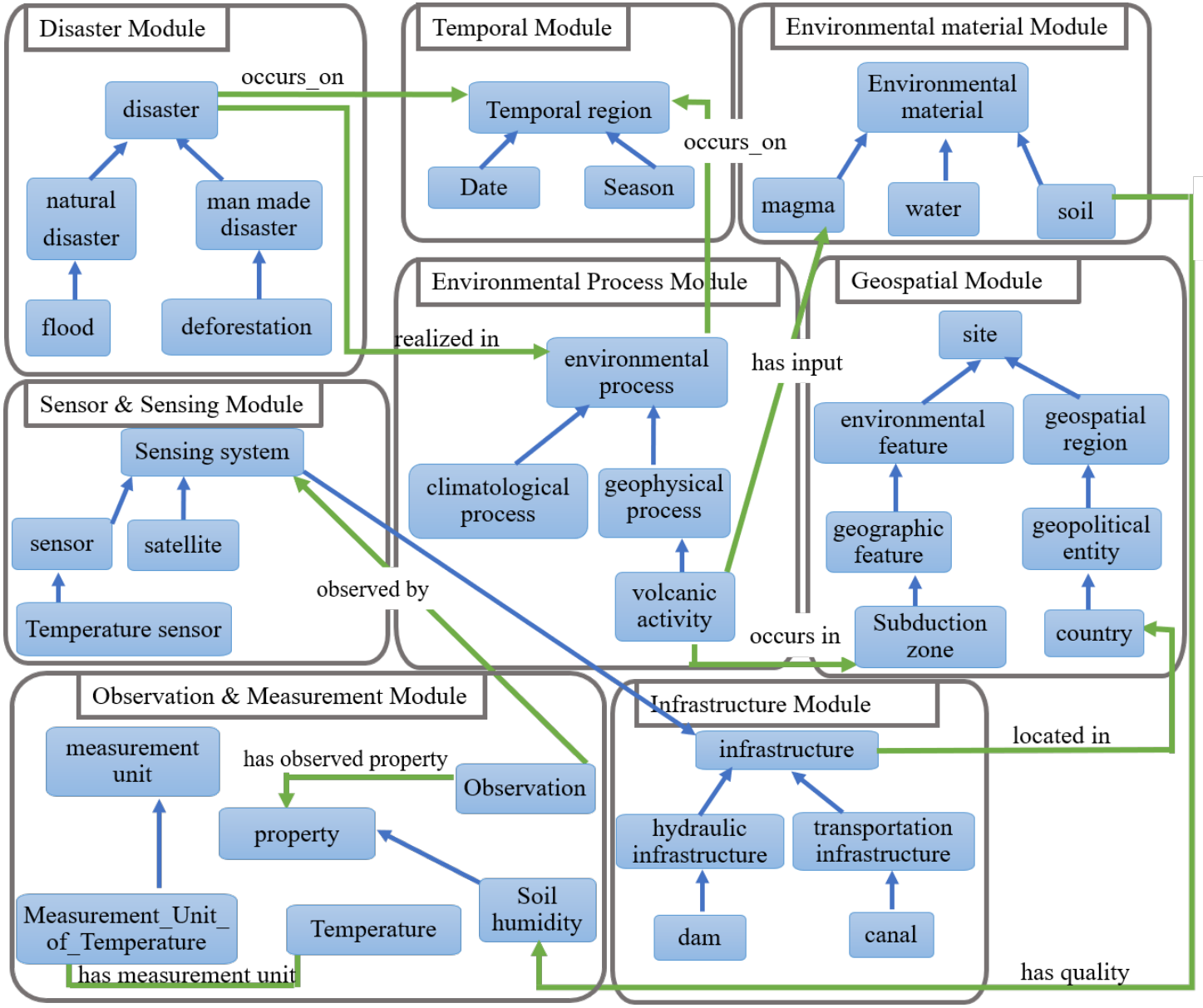


Partial View of some modules





Partial view of the integration (interrelation) of modules



The whole inetgrated ontology includes more than 1200

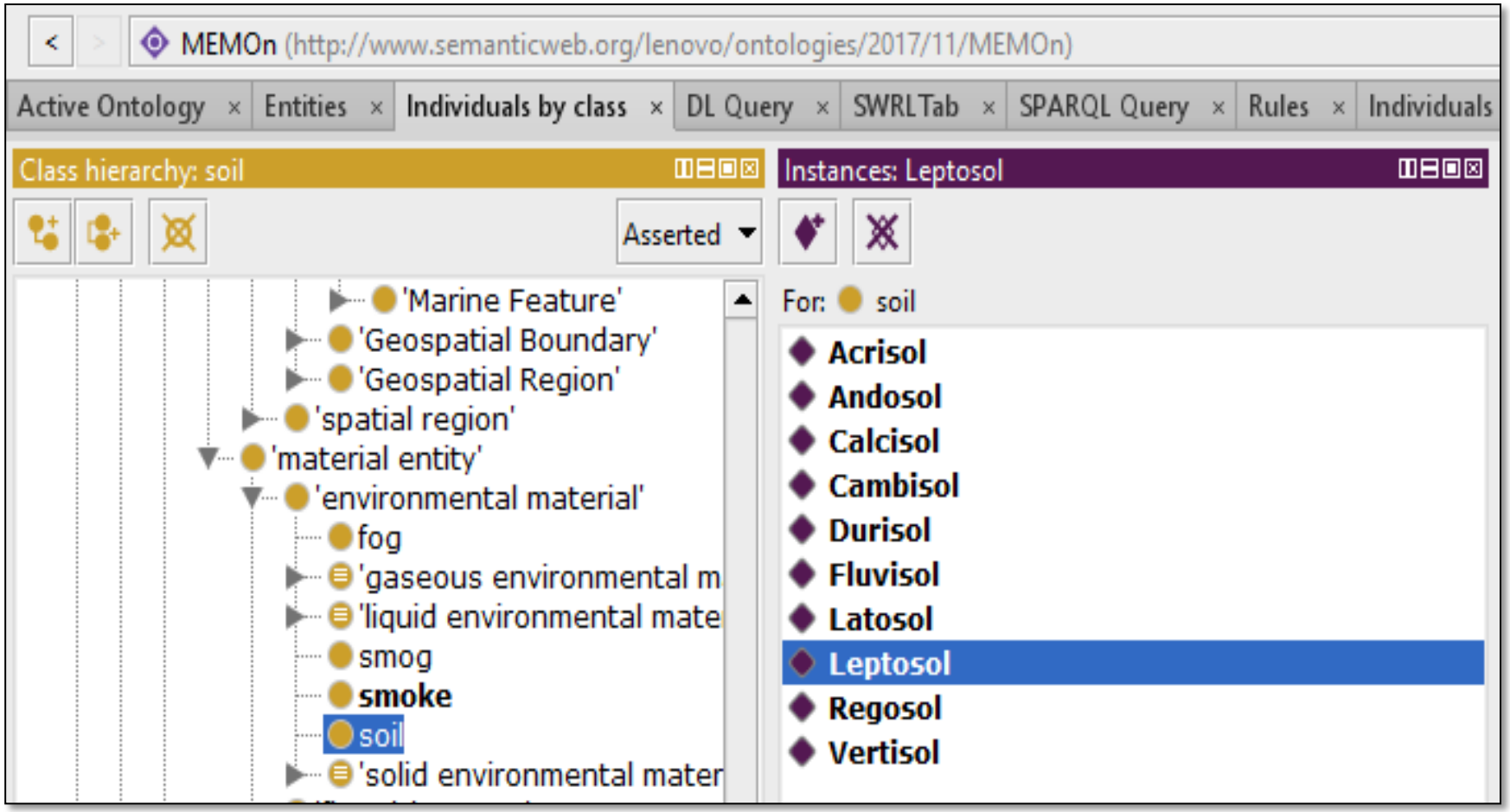
Definition of rules

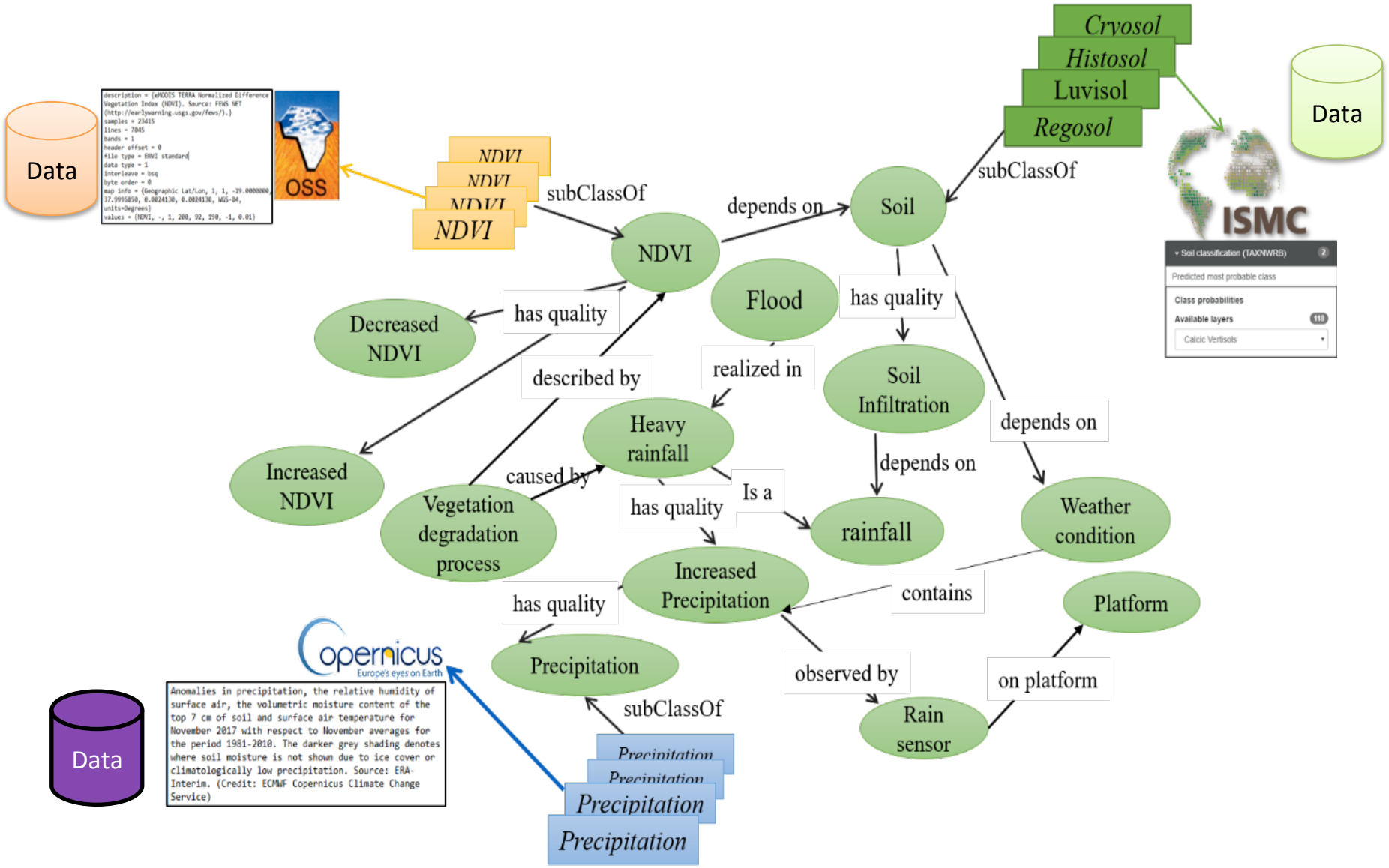
	Purpose	SWRL rule
R1	Classify an environmental process to a specific type of process.	‘water-based rainfall’(?r), precipitation(?p), 'has precipitation value'(?r, ?p), swrlb:greaterThan(?p, 16), swrlb:lessThan(?p, 50) -> 'very heavy rainfall'(?r)
R2	Relate flooding process to the cause of very heavy rain.	'very heavy rainfall'(?r), date(?d), 'spatial region'(?rg), 'occurs on'(?r, ?d), 'occurs at'(?r, ?rg), flooding(?f), 'occurs on'(?f, ?d), 'occurs at'(?f, ?rg) -> 'caused by'(?f, ?r)
R3	Relate a soil type to limited infiltration.	soil(Regosol), 'located in' (Regosol, ?rg), 'spatial region'(?rg), 'soil infiltration' (?inf) -> 'limited soil infiltration'(?inf)
R4	Reclassify a general disaster to a specific type of disaster.	flood(?f), date(?d), 'spatial region'(?rg), 'occurs on'(?f, ?d), 'occurs at'(?f, ?rg), 'water runoff'(?ro), 'heavy runoff'(?ro), 'occurs on'(?ro, ?d), 'occurs at'(?ro, ?rg) -> 'riverine flood'(?f)

Examples of queries on the model

CQ	SPARQL Query	Results
Q1: What are the environmental processes which can cause a flood?	SELECT ?process WHERE { ?process a envo:'hydrological process' . ?f a envo:flood .?f ero:'caused by' ?process }	'dam failure' 'heavy rainfall' 'water runoff'
Q2: What are the spatial zones where a volcanic activity is the highest?	SELECT ?region WHERE { ?region a obo:site .?v a memon:'volcanic activity' . ?v obo:'occur in' ?region .?q a memon:'volcanic exposed' . ?region ero:'has quality' ?q }	'mid-ocean ridge' 'subduction zone'
Q3: What are the environmental materials involved in effusive volcanic eruption?	SELECT ?material WHERE { ?material a envo:'environmental material' . ?v a envo:'effusive eruption'. ?vero:'has input' ?material }	magma
Q4: What is the type of sensor used for ground vibration?	SELECT ?s WHERE { ?s a ao:sensor.?s memon:observes ?obs ?obs a memon:'ground vibration' }	Seismometer
Q5: What are the properties observed and associated to the heavy rainfall?	SELECT ?obs WHERE { ?w a memon:'weather condition of rainfall' . ?w ro:contains ?obs . ?hr memon:described_by ?w . ?hr a memon:'heavy rainfall' }	precipitation wind humidity
Q6: What is the disaster caused by a heavy rainfall?	SELECT ?d WHERE { ?d a memon:disaster. ?d ero:'disposition_of' ?hv. ?hv a memon:'very heavy rainfall' }	flood
Q7: What unit is pressure measured in?	SELECT ?u WHERE { ?u a info: 'measurement unit'. ?p a obo:pressure . ?p memon:'has measurement unit' ?u }	bar

Instantiation of the Knowledge base





Linking multi-source data with the Ontology.

Classes are shown as rounded boxes, individuals as rectangles.

Queries on the KB vs Databases

Query	NOAA Database	OSS Database	ISMC Database	MEMOn
Query A: select average annual precipitation where soil type is "Vertisol"	No result	No result	No result	Between 500 and 1000 mm
Query B: Select soil type in arid climate	No result	No result	No result	Arenosol
Query C: select environmental process that may occur where precipitation equal to 30mm/h	No result	No result	No result	heavy rainfall, soil infiltration, vegetation degradation process